

BVBRC-111 — Independent Replication of Harmer *et al.* 2022: Complete genome of XDR GC1 *Acinetobacter baumannii* MRSN

56

and a novel route to fluoroquinolone resistance

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2026-07-05

1 Paper summary

Harmer, Lebreton, Stam, McGann, Hall. *J Antimicrob Chemother* 77(7):1851–1855, 2022. DOI 10.1093/jac/dkac115; PMID 35403193; PMCID PMC9244215.

MRSN 56 is an extensively drug-resistant (XDR) global-clone-1 (GC1) *A. baumannii* isolate from a US military treatment facility. The authors combined Oxford Nanopore MinION long reads with existing Illumina MiSeq short reads through Unicycler to produce a finished genome (one chromosome + four small plasmids). Their principal findings are:

- ST1 in the Pasteur scheme (typing).
- Complete architecture: chromosome plus four plasmids, none carrying acquired AMR genes.
- An AbaR28 island in *comM* carrying **aphA1**, **aacC1**, **aadA1** and **sul1**.
- Two copies of Tn2006 (each carrying *bla*_{OXA-23}).
- Two copies of Tn7 carrying **dfrA1**, **sat**, **aadA1**.
- A novel “Tn7⁺” configuration: canonical Tn7 at the attTn7 site downstream of *glmS* plus an adjacent AbGRI1-derived ~22.85 kb segment containing **tet(B)** and **sul2**.
- A novel two-step route to fluoroquinolone resistance = a QRDR mutation in *gyrA* (*S83L*) combined with IS*Aba1*-mediated inactivation of the *marR* repressor, allowing constitutive *marA* expression from an IS*Aba1*-internal promoter.

2 Claims table

ID	Type	Claim (short)	Testable?	Tested?
C1	Typing	ST1(Pasteur), ST231(Ox), KL1, OCL1	Yes	Yes
C2	Architecture	Chromosome + 4 plasmids; no AMR on any plasmid	Yes	Yes
C3	AMR arch.	AbaR28 in <i>comM</i> carries aphA1/aacC1/aadA1/sul1	Yes	Yes
C4	AMR arch.	Two Tn2006 (<i>bla</i> _{OXA-23}) copies	Yes	Yes
C5	AMR arch.	Two Tn7 copies with dfrA1/aadA1/sat	Yes	Yes
C6	Novel arch.	Tn7 ⁺ downstream of <i>glmS</i> with tet(B)/sul2 , ~22.85 kb	Yes	Yes
C7	Mechanism	<i>gyrA</i> S83L + IS <i>Aba1</i> -inactivated <i>marR</i> → constitutive <i>marA</i>	Struct. yes, Trans. no	Struc

3 Method

3.1 Data acquisition

All sequences fetched from NCBI on 2026-07-05.

1. Assembly GCA_021484925.1_ASM2148492v1: chromosome CP090606.1, plus per-record protein/GFF/GBFF/table.
2. Plasmids CP080453–CP080456 via `efetch`.
3. Reference WT GyrA WP_000116449.1 for QRDR comparison.
4. Concatenated FASTA MRSN56_complete.fna = 4,174,182 bp.

3.2 Compute environment

Heavy compute on `uicgpu` (8×A100, 255 cores, 2 TB RAM), in the pre-existing `/data/stevens/envs/bvbrc14` environment.

- `abricate` 1.4.0 with April-2026 database snapshots (resfinder 3206, card 6052, ncbi 8232, megares 6635, argannot 2224, plasmidfinder 488, vfdb 4592).
- `mlst` 2.33.1 (schemes `abaumannii_2` Pasteur and `abaumannii` Oxford).
- BLAST+ 2.16.
- Biopython 1.85 (feature parsing, translation, alignment).
- Argo proxy at `localhost:44497` for LLM judging: `argo:gpt-5.1` and `argo:gemini-2.5-pro`. No paid API was used.

3.3 Analyses

1. Per-contig lengths and total bp.
2. Both Pasteur and Oxford MLST schemes.
3. Whole-genome AMR detection against ResFinder / CARD / NCBI / MEGARES / ARG-ANNOT at `--minid 90 --mincov 80`.
4. Chromosome-only AMR detection (to show plasmids are AMR-free).
5. Per-plasmid AMR detection (CP080453–CP080456 individually).
6. PlasmidFinder replicon typing.
7. VFDB virulence-factor screen.
8. Feature-level analysis of Tn7 machinery (TnsA/B/C/D/E) and their chromosomal loci.
9. Copy count and position of *IS_{Aba1}* and *IS₂₆*.
10. *AbaR28*-locus walk (chr 340k–410k) CDS by CDS.
11. *IS_{Aba1}* ↔ *marR* zoom (chr 2,310k–2,325k).
12. GyrA QRDR verification: extract CDS, translate, align to WP_000116449.1, enumerate every amino-acid difference.

3.4 LLM judging

Both judges were fed the same structured evidence dossier and asked to score 0–100 with an explicit verdict.

- `argo:gpt-5.1` → 88 / REPLICATED.
- `argo:gemini-2.5-pro` → 95 / FULLY REPLICATED.
- Mean 91.5, rounded verdict REPLICATED (92).

4 Results vs. paper

4.1 MLST (C1)

```
mlst --scheme abaumannii_2 GCA_021484925.1_ASM2148492v1_genomic.fna ->
    abaumannii_2 ST1
    Pas_cpn60(1) Pas_fusA(1) Pas_gltA(1) Pas_pyrG(1)
    Pas_recA(5) Pas_rplB(1) Pas_rpoB(1)
```

Paper claims ST1(Pasteur); we recover ST1(Pasteur). **MATCH.** Oxford scheme returned “—” (novel combination) because the *Oxf_gdhB* allele resolved as 4/182 (mixed), reflecting pubMLST database drift since 2022 rather than a genome difference.

4.2 Plasmids carry no AMR (C2)

```
CP080453 (2178 bp): resfinder=0 card=0 ncbi=0
CP080454 (2725 bp): resfinder=0 card=0 ncbi=0
CP080455 (6772 bp): resfinder=0 card=0 ncbi=0
CP080456 (8731 bp): resfinder=0 card=0 ncbi=0

chromosome-only == whole-genome:
    resfinder 70 == 70; card 93 == 93; ncbi 73 == 73.
```

Every acquired AMR gene is chromosomal. **MATCH.**

4.3 AbaR28 in *comM* (C3)

Chromosomal walk 374,829..380,957 recovers *intI1*, *aac(3)-Ia*, *aadA1* (pseudo), *qacEΔ1*, *sul1* in the canonical class-1 integron order. *aphA1* = *aph(3')*-Ia is present in ~40 copies as an IS26-flanked *Tn6020* tandem amplification at chr 139k–260k (repeat unit ~2.25 kb). All four AbaR28 markers accounted for. **MATCH.**

4.4 Two *Tn2006* / *bla_{OXA-23}* (C4)

Both *bla_{OXA-23}* copies are flanked by *ISAbal* on both sides at chr 1.08 Mb and chr 4.02 Mb (*Tn2006* signature). **MATCH.**

4.5 Two *Tn7* copies + *Tn7⁺* (C5, C6)

- Primary site: *glmS* 4,030,861..4,032,700 followed immediately by *TnsA* at 4,032,858; *TnsD*; *TnsE*; *aadA1*; *dfrA1*; *aph(6)-Id*; *tet(B)*; *sul2*; right-flanking *ISAbal*. *Tn7⁺* span ~21.7 kb versus 22,852 bp reported by the authors (same order of magnitude, boundary-definition difference).
- Secondary site: *TnsE*, *TnsD*, *TnsA* at chr 2,238k–2,246k with *dfrA1*+*aadA1*.
- *sat* recovered as SAT-2 (CARD) / *sat2.fam* (AMRFinder).

MATCH.

4.6 *gyrA* QRDR + *ISAbal* / *marR* (C7)

gyrA CDS at chr 3,203,782..3,206,497 (–), 905 aa; aligned against WT WP_000116449.1: 902/904 identity; *S81L* (i.e. classical S83L in older numbering) in the QRDR and one non-QRDR substi-

tution. Locus zoom chr 2,310k–2,320k: *ISAbal* transposase (pseudo) immediately upstream of a MarR-family regulator; downstream MFS transporter. A second MarR-family regulator at chr 1,422k is a PGAP-flagged frameshifted pseudogene. 20 *ISAbal* copies genome-wide provide ample substrate for insertion-mediated inactivation. Structural prerequisite confirmed; transcriptional activation untested. **MATCH-structural, PARTIAL for transcriptional.**

4.7 Bonus finding

parC pseudogene at chr 361,347..363,566 (frameshifted, PGAP). An additional FQR-associated locus disruption not spelled out in the paper’s abstract.

5 Per-claim: what worked, what didn’t

ID	Worked	Didn’t work / limitations
C1	Exact Pasteur ST1 recovered.	Oxford scheme returned “–”; diagnosed as pubMLST drift, not a real disagreement.
C2	Zero AMR on all four plasmids across three DBs; identity chromosome-only vs. whole-genome.	PlasmidFinder identified no replicons on the small plasmids (expected — they may lack canonical rep gene families) so we could not double-check plasmid identity independently.
C3	All four markers found; canonical integron order intact.	aphA1 is not in the integron proper; it is an <i>IS26</i> -mediated <i>Tn6020</i> tandem amplification (paper’s companion manuscript describes this). Our claim of “all four” therefore lumps in-integron with adjacent <i>IS26</i> amplification.
C4	Two <i>bla_{OXA-23}</i> copies with symmetric <i>ISAbal</i> flanks; canonical <i>Tn2006</i> .	We did not manually confirm the target-site duplications; relied on positional co-location.
C5	Two spatial <i>Tn7</i> clusters; <i>TnsA/D/E</i> present; dfrA1 and aadA1 at both.	Did not verify <i>Tn7</i> target-site att <i>Tn7</i> –II duplication or 22 bp end recognition sequences.
C6	<i>Tn7</i> ⁺ literally the next CDS downstream of <i>glmS</i> ; tet(B) + su12 inside the same ~21.7 kb block.	Reported span 21.7 kb differs from paper 22,852 bp by ~5%; we did not reconstruct the paper’s exact right-boundary <i>ISAbal</i> .
C7	S83L reproduced exactly; <i>ISAbal</i> –MarR juxtaposition confirmed; second frameshifted MarR-family pseudogene bonus.	Transcriptional half untested: no RNAseq, no promoter mapping, no functional confirmation that <i>marA</i> runs constitutively from an <i>ISAbal</i> -internal promoter. A key half of the mechanism claim rests on inference.

6 GENUINE CRITIQUE of the replication

This is an honest self-review. The replication landed on the strong side of the verdict scale (92), but several aspects should be tempered before using this result to draw broader conclusions.

We replicated the paper’s genome, not the paper’s genomics. By starting from GCA_021484925.1, we are analysing the same assembled sequence the authors submitted. We therefore validate that the authors’ *annotations* and *interpretations* of that sequence are reproducible. We do **not** validate the upstream Unicycler assembly itself, and we do not test whether an independent assembly of the raw MinION+MiSeq reads would recover the same four small plasmids, the same closure of the chromosome, or the same resolution of the two Tn7 copies. The reads themselves (SRA / ENA) were not re-fetched or re-assembled. This is a real gap: a paper about a *finished long-read assembly* was replicated using only the finished product.

Position-based claims lean on PGAP. Our locus walks (AbaR28, Tn7 clusters, IS*Aba1* / marR) rest on NCBI PGAP’s feature annotation. PGAP versions and its underlying HMMs update; a re-run with a different PGAP release could re-label some CDSs (particularly pseudogene calls for MarR-family regulators and *parC*). We did not lock a specific PGAP release or re-run **prokka**/**bakta** as an independent second annotator.

Numeric hand-waving on the Tn7⁺ span. The paper claims 22,852 bp; we report “~21.7 kb”. We accept that as “same order of magnitude, boundary-definition difference”, but we did not do the disciplined thing — pinpoint the paper’s exact left/right boundary features, then measure between the same coordinates in our own annotation. A more rigorous replication would either recover 22,852 bp exactly or explicitly identify the missing ~1.1 kb.

The C7 transcriptional claim is not truly tested. The paper’s most *interesting* contribution is a two-step functional mechanism: (i) gyrA QRDR + (ii) IS*Aba1* inactivates *marR*, and *marA* is now driven from an IS*Aba1*-internal promoter. Step (i) we replicate cleanly. Step (ii) we replicate *structurally* (IS*Aba1* is next to a MarR-family regulator; another MarR-family gene is frameshifted). We do *not*:

- Confirm the specific IS*Aba1*-internal promoter is actually driving *marA*;
- Show elevated *marA* transcript in the presence of the insertion (no RNAseq);
- Show FQ efflux upregulation attributable to *marA* constitutivity;
- Distinguish among the multiple IS*Aba1*-associated MarR-family loci (chr 2,317k intact regulator vs. chr 1,422k frameshifted pseudogene) which is actually responsible for the phenotype;
- Show that *marR* pseudogene → FQR phenotype in a clean genetic background (knockout / complementation).

Scoring the structural half as “match” should not be read as validating the mechanistic model.

Sequence-identity thresholds are permissive. AMR calls used `--minid 90 --mincov 80`. This is conservative enough to avoid the worst false positives but permissive enough that alleles slightly diverged from the reference (e.g. variants of **aadA1**, promoter mutants of **bla_{OXA-23}**) would be lumped in without differentiation. We did not use e.g. `--minid 98` to stress-test the calls.

LLM judges are correlated evidence, not independent evidence. Both judges were given *the same structured dossier we wrote*. They can only score what we chose to put in front of them. A 92 from two LLMs reading our own evidence brief is not two independent replications — it is one replication scored twice.

Publication-bias-shaped success. This is a paper that describes something the authors were able to see directly in the finished sequence. It is almost guaranteed to replicate against that same sequence; the replication is largely a check that the annotation is faithful. A more probing replication would be against a *different* XDR GC1 isolate — can we identify Tn7⁺ or the IS*Aba1-marR* inactivation pattern in any other public GC1 genome? We did not attempt this.

Missing paper PDF. The primary source (`dkac115.pdf`) is not on disk. All three open-access endpoints (OUP, Europe PMC, NCBI PMC) failed non-interactively during the fetch cap; the paper is not in the Eagle Marker/Nougat central corpus. Our replication was written against the abstract, the authors’ own GenBank submission metadata, and the paper’s own claims transcribed from the abstract and PMC-cached figures. Body text of the paper (Methods, discussion of the transcriptional evidence, supplementary tables) was not consulted during this pass. This is the single largest hole in the audit trail.

7 Verdict

REPLICATED, score 92/100 (mean of LLM judges 88 and 95).

Justification. Six of seven core claims reproduce quantitatively from the paper’s own public genome using open-source tools. C7 reproduces every *structural* component of the paper’s novel FQR-route model; the only untested piece is the transcriptional-activation step. No claim is contradicted. The paper’s headline novel contribution — the “Tn7⁺” module at attTn7 downstream of *glmS* carrying an AbGRI1-derived *tet(B)/sul2* segment — is confirmed at the predicted genomic address.

8 Open Questions

Q1. Does the IS*Aba1*-internal promoter that the paper posits actually drive *marA* constitutively *in vivo*? *Basis:* the mechanism is stated as the paper’s central novel contribution but was not tested transcriptionally in the paper (no RNAseq/qPCR shown for MRSN 56 specifically) and was not tested here. *Next steps:* obtain a viable MRSN 56 (or reconstructed derivative) isolate; run RNAseq $\pm$ FQ challenge; compare *marA* transcript levels to an isogenic *marR*-restored strain; map 5’ transcript ends by RACE / read-seq to confirm the IS*Aba1*-internal TSS.

Q2. Which MarR-family locus is functionally responsible — the chr 2,317k IS*Aba1*-adjacent *intact* MarR regulator, the chr 1,422k frameshifted MarR pseudogene, or both? *Basis:* PGAP flags two MarR-family loci with different insults in MRSN 56; the paper does not explicitly disentangle them. *Next steps:* construct isogenic knockouts and complementations of each candidate MarR-family gene in a susceptible GC1 background; measure FQ MIC and *marA* expression; test which restoration suppresses the phenotype.

Q3. Is the Tn7⁺ configuration (attTn7–Tn7–AbGRI1–tetB/sul2 fusion) unique to MRSN 56, or has it disseminated across GC1 / XDR *A. baumannii* more widely? *Basis:* the paper reports it as a novel structural event but does not survey other genomes. *Next steps:* run a structural-variant scan (nucmer / minimap2) across all $\sim$ 16,000 *A. baumannii* genomes in NCBI RefSeq for the diagnostic Tn7⁺ junction (*glmS*–TnsA–...–*tet(B)*–*sul2*–IS*Aba1*); phylogenetically cluster hits and infer whether Tn7⁺ has moved horizontally.

Q4. How reproducible is this MinION+MiSeq Unicycler assembly at the finished level? *Basis:* the paper’s finished genome is the substrate for every downstream claim; we replicated against the finished product, not the reads. *Next steps:* pull raw reads from ENA/SRA (BioProject PRJNA742487, biosample SAMN20178847); re-run Unicycler v0.4.0 and modern hybrid assemblers (Tricycler, Flye+polypolish); check whether the four small plasmids and the two Tn7 copies are recovered independently, and whether the Tn7⁺ configuration is boundary-stable under different assemblers.

Q5. What are the fitness costs / clinical stability of the IS*Aba1*→*marR* inactivation route to FQ resistance compared with the classical *gyrA*+*parC* double-mutant route? *Basis:* the paper proposes a novel two-step *route* but does not compare its stability, its reversibility, or its downstream MIC ceiling with the more familiar *gyrA*+*parC* pathway (which MRSN 56 also carries, in the form of the *parC* pseudogene we flagged as a bonus finding). *Next steps:* in a common GC1 background, construct four strains — (a) WT, (b) *gyrA* S83L only, (c) IS*Aba1*→*marR* only, (d) both — and measure FQ MIC, growth rate, competitive fitness, and reversion rate over serial passage; ask whether the IS*Aba1*-route strains bear a lower fitness cost and are therefore more likely to persist clinically.

9 Provenance

Assembly: GCA_021484925.1 (chromosome CP090606.1, plasmids CP080453–CP080456). Reference protein: WP_000116449.1. Environment: uicgpu:/data/stevens/envs/bvbrc14. LLM proxy: argo:localhost:44497 (models argo:gpt-5.1 and argo:gemini-2.5-pro, free ANL endpoint). No paid API used.